The role of population size and density in evolutionary rescue

Rescue Crew

Summary to-date from theory meetings

Setting up the problem:

Classical rescue theory poses that increased population size monotonically increases the probability of rescue (CITE-G/H, O/U), because larger population sizes help buffer the population from decline (ecological effect), are generally associated with higher genetic variance (from mutation-selection balance) and higher mutation accumulation (more absolute reproduction). Specifically, the dependence has been known to be negative-exponential

$$P_{\text{res}} = 1 - \exp(-\theta N_0) \tag{1}$$

For instance, in a model with two alleles (mutant and wildtype) where the frequency of mutants p_{sgv} is constant, the wildtype declines exponentially at rate $1 - r < 1$ and establishment probability p_{est} doesn't depend on the population size, we have $\theta = p_{\text{sgv}} + p_{\text{est}}\mu\frac{N_0}{r}$. This negative exponential dependence on N_0 also holds for a variety of other cases such as for a quantitative trait, ...

Given this clear dependence and the ease of measuring population size in experiments and the field, it is one of the better-known and useful(?) relationships. Experimental work has corroborated the qualitative dependence that increasing population size increases the probability of rescue (CITE). However, quantitative tests are few. (EXPERIMENTAL GROUP info on quantitative tests)

Further, θ could depend on population size in a variety of cases. For instance, if the wildtype fitness is density dependent, then it can affect the mutational input while density dependence in mutant fitness can alter its establishment probability. Density dependence could be negative (such as due to competition) or positive (such as due to Allee effects). In addition other parameters, such as the frequency of mutants in the SGV could also depend on N_0. For instance if N_0 is also the size of the population before the environment changes, it can also affect the p_{sgv} due to drift.

It is important to note that the dependence of theta on population (census) size could be via its density i.e., the number of individuals per unit resource (such as area, food etc.). For instance, the fitness dependence (of the mutant or wildtype) of the probability of rescue is most likely due to density rather than census size, i.e. we would expect competition (a source of negative-density dependence) to be unchanged if we increase the number of individuals but

also the amount of available resources. On the other hand, the frequency of mutants in the SGV changes due to the the magnitude of drift which depends on the population size before change. Therefore, it is important to distinguish between this dependence on density and population size, especially when proposing experiments.

In this work, (1) we explore the dependence on θ on population density and size with a focus on distinguishing between the two, (2) use simulations to test if and how this dependence can be statistically detected from data, and (3) propose experiments to do test these. By jointly analysing theory and experimental studies, we provide key insights into a key gap in the evolutionary rescue literature.

Results

Here we explore two models where θ is density-dependent, specifically via fitness of the wildtype and mutant to make the following salient points

1. Population size dependence can be detected from rescue probabilities
2. However, smaller or sparser range of population sizes makes it harder especially for positive-density dependence
3. Despite density dependence fitness of θ, the probability of rescue might have the canonical negative exponential dependence on N_0 if the density is kept constant, i.e. $\frac{N_0}{K}$ is constant.

Negative density dependence

We explore a continuous-time model where birth rates of mutants and wildtypes have negative-density dependence

$$b_i(N) \equiv b_i \times (1 - \frac{N_m + N_w}{K}) \; ; \; i = w, m$$

while death rates are constant $d_i; i = w, m$. We also denote $r_i = b_i - d_i$ to denote the mean growth rate in the absence of density dependence. An important thing to note is that the "Carrying Capacity" (defined as the non-zero equilibrium of the deterministic model) for each type is different, $K_i = K\frac{r_i}{b_i}$.

We start with a population with no standing genetic variation with N_0 wildtypes. Mutants are generated via mutations at rate μ and a mutant generated at time t establishes with probability $p_{\text{est}}(t)$. The rescue probability can be computed by considering a deterministic decline of the wildtype and computing P_{res} as in Uecker and Hermisson (2011). We provide a closed-form expression for $p_{\text{est}}(t)$ in the appendix. Here we show the final form,

$$P_{\text{res}} = 1 - \exp(-\mu \int_0^\infty N_w(t) p_{\text{est}}(t)) \tag{2}$$

$$= 1 - \exp(-\mu \frac{K}{b_w} \int_0^{\frac{n_0}{n_0 - \frac{r_w}{b_w}}} \frac{p_{\text{est}}(x)}{1-x} dx), \tag{3}$$

where $n_0 = \frac{N_0}{K}$. The second equality was obtained with a change of variable from t to density $x = \frac{n_0 e^{r_w t}}{n_0 - \frac{r_w}{b_w}}$. Thus, the rescue probability only depends on the initial density n_0 and not on the population size. Further, unlike the classical case, the exponent doens't necessarily go to infinity as $n_0 \to \infty$.

We then ran simulations for cases with no competition ($K \to \infty$), and two degrees of competition ($K = 10^4, 3 \times 10^4$). We then fit each of these to to a negative exponential function (assuming θ is density independent). We find that with the measurement of rescue probability across a sufficient range of population, one can detect effects of competition, even with low number of replicates (compare Fig. 1 a and c). However, lower range (Fig. 1 b and e) and sparser data (Fig. 1 c and f) can make it harder to detect these trends. Further, if we also increase the

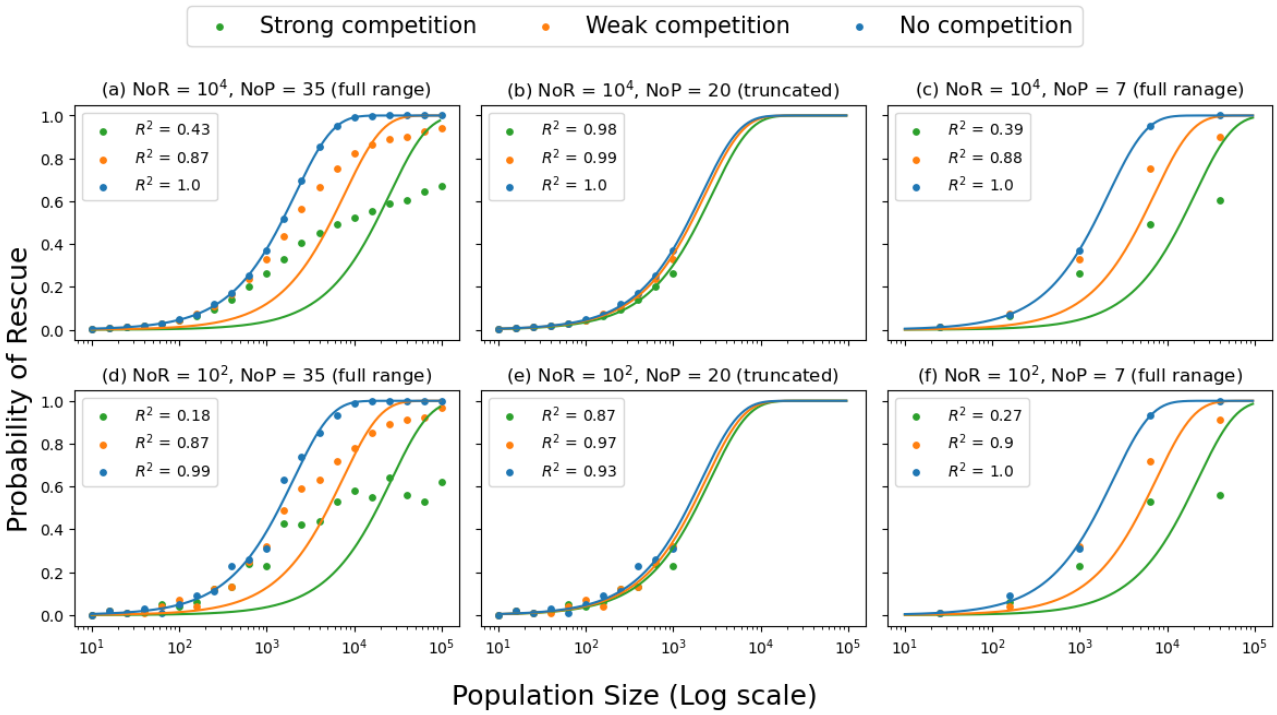

Figure 1: Probability of rescue in a birth-death model with no density dependence ($K = \infty$; blue lines), weak ($K = 3 \times 10^4$; orange lines) and strong negative ($K = 10^4$; green lines) density-dependent birth rates. NoR is the number of replicates and NoP is the number of points for each line. The fitted line is an MLE under a GLM with Binomial distribution and link function $-\log(1-p)$ as in Martin *et al.* (2013). The goodness of fit measure here is an R^2 value (there are prob other better options. Martin *et al.* (2013) uses a χ^2 test on something called Deviance but idk what that is atm). Parameters : $b_w = \frac{1+\ln(0.95)}{2}, d_w = \frac{1-\ln(0.95)}{2}, b_m = \frac{1+\ln(1.15)}{2}, d_w = \frac{1-\ln(0.15)}{2}, \mu = 10^{-4}$.

carrying capacity with initial populations size, we have an exact negative exponential depence on N_0 (See Eq. 3 when $K \propto N_0$). We see this in simulations as well (Fig. 2).

Positive density dependence fitness

Here, we consider a model where there is positive-density dependence in fitness. Specifically, we consider a model with mate-finding Allee effect (as in **?**). This is a discrete time mode with non-overlapping generations with hermaphrodites where the probability of the female gamete being fertilized is given by $1 - \exp(-a(N-1))$ where N is the population size. Individuals produce offsprings according to a Poisson distribution with mean $\lambda_i; i = w, m$. Offspring inherits

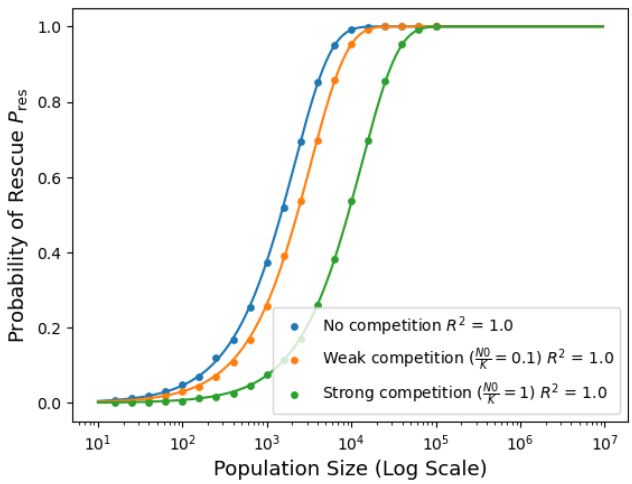

Figure 2: Rescue probability plotted against initial population size, N_0, when carrying capacity is scaled proportional to N_0. We do this for different degree of competition determined by $\frac{N_0}{K}$.

the trait of either parent with equal probability. We provide the model details in the appendix. Since we build the model mechanistically, the "Allee threshold" (defined as the population size below which extinction is guaranteed in a monomorphic population) is different for the two types and more importantly dependends on the mean offspring number. For the wildtype it is ∞ (since it is subcritical) and is given by $a_i = -\frac{1}{a}\ln(1 - \frac{1}{\lambda_m})$

However, the key point is that it is hard to detect positive density-dependence, even when we have high replicates and large range of population sizes (Fig. 3 a and d). However, fitting the model by excluding the saturated part (where $P_{\text{rescue}} = 1$) helps in distinguishing the two much better (Fig. 3 b and e).

Distinguishing between positive and negative density

Note, that the statistics used in the previous sections just detect (when possible) if the rescue probability follows the negative exponential or not. However there are some clear differences between the positive and negative density dependence plots. The main one being that the simulated values the behavior of the difference between the simulated values and the model fit. We plot this in Fig. 4. For positive density dependence (the left figure) the difference goes from negative to positive while for competition (right figure) it is the other way round. Further, for negative-density dependence the saturating values need not be 1 so the difference doesn't go to 0.

Comments and next steps:

1. Consider the statistical techniques a bit more carefully. Given, a key aim is to suggest experiments, it would be useful to spend some time on the statistics of it all. Something along formalizing the patterns of Fig. 4.

2. Make connection with the "7 types of rarity" paper. Consider how to best frame this, e.g., do we want to show the seven types across a census/density plot and then highlight the 2-3 examples we simulate or allude to which "sections" of the plot might differ from classical expectations, e.g., high density/low population size. Puneeth's current

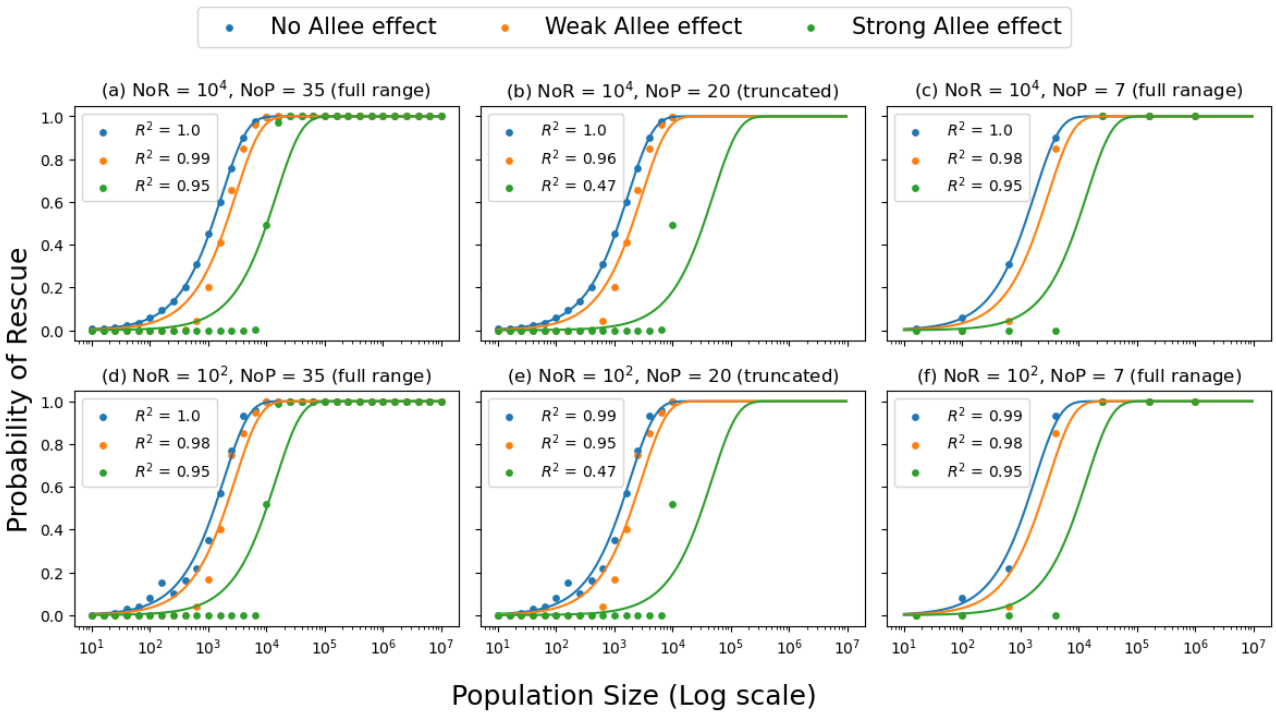

Figure 3: Probability of rescue in a discrete model without density dependence ($a = \infty$; blue lines) and with weak ($a = 0.01$; orange lines) and strong ($a = 0.001$; green lines) mate-finding Allee effect. NoR is the number of replicates and NoP is the number of points for each line. The fitted line is an MLE under a GLM with Binomial distribution and link function $-\log(1-p)$ as in Martin *et al.* (2013). The goodness of fit measure here is an R^2 value (there are prob other better options. Martin *et al.* (2013) uses a χ^2 test on something called Deviance but idk what that is atm). Other parameters: $\lambda_w = 0.95, \lambda_m = 1.5, \mu = 10^{-4}$.

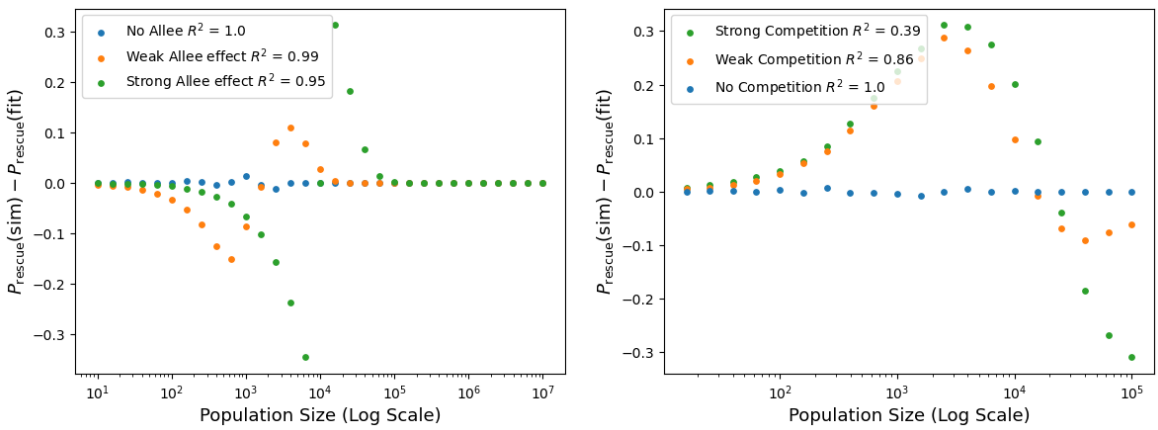

Figure 4: The difference between the rescue probability obtained via simulations. $P_{\text{res}}(\text{sim})$ and the one obtained from fitting the negative exponential model with a constant θ, $P_{\text{res}}(\text{fit})$ for the model with positive (left) and negative (right) density dependence. The parameter values for simulations are given in Figs. 3 and 1, respectively.

simulations address the first of these questions; I'm not sure about "measuring" the density directly but it's something that I believe we've talked about a touch in our meetings. I think the second question (about metrics rather than processes) would be try to make sure we hit different sections of the "types of rarity", i.e, does density "act" truly independently of population size or is there some interaction there? Four examples could be (density/census): low/small, low/large, high/small, high/large.

3. We should consider Tanaka and Wahl (2022) paper since it already has a plot of P_{res} as a function of N_0 and the dependence is via size as opposed to density. Potentially, could connect to Laure's experimental work?

A Birth-death process with competition - Model details

B Mate-finding Allee Effect - Model details